INCH-POUND
MIL-DTL-18045G
1 July 1997
SUPERSEDING
MIL-L-18045F
8 August 1986
(See 6.6)

DETAIL SPECIFICATION

LIFE PRESERVERS, VEST, INHERENTLY BUOYANT

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers inherently buoyant, vest type, life preservers which use buoyant foam pads.

1.2 Classification. Life preservers will be of the following types, as specified (see 6.2).

Type I - Adult's (with collar)
Type II - Adult's (without collar)

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, SEA 03R42, Naval Sea Systems Command, 2531 Jefferson Davis Hwy, Arlington, VA 22242-5160 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

A-A-52071 - Tape, Textile, Cotton, General Purpose.
V-F-106 - Fastener, Slide, Interlocking.
V-T-285 - Thread, Polyester.
V-T-295 - Thread, Nylon.
CCC-C-426 - Cloth, Drill, Cotton.
DDD-L-20 - Label: For Clothing, Equipage, and Tentage, (General Use).

DEPARTMENT OF DEFENSE

MIL-P-15280 - Plastic Material, Unicellular (Sheets and Tubes).
MIL-W-17337 - Webbing, Textile, Woven Nylon.
MIL-A-24179 - Adhesive, Flexible Unicellular-Plastic Thermal Insulation.

STANDARDS

FEDERAL

FED-STD-191 - Federal Standard for Textile Test Methods.
FED-STD-595 - Colors Used in Government Procurement.
FED-STD-751 - Stitches, Seams and Stitchings.

(Unless otherwise indicated, copies of the above specifications, standards, and handbooks are available from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

PUBLICATIONS

DEPARTMENT OF TRANSPORTATION (DOT)

46 CFR 164.018 - Retroreflective Materials for Lifesaving Equipment.

(The Code of Federal Regulations (CFR) and the Federal Register (FR) are for sale on a subscription basis by the Superintendent of Documents, US Government Printing Office, Washington, DC 20402. When indicated, reprints of certain regulations may be obtained from the Federal Agency responsible for issuance thereof.)

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DRAWINGS

NAVAL SEA SYSTEMS COMMAND (NAVSEA)

NAVSHIPS 803-1419479 - Life Preserver, Vest, Inherently Buoyant: Assembly.
NAVSHIPS 803-1419480 - Life Preserver, Vest, Inherently Buoyant: Patterns for Inside and Outside Jacket Section.
NAVSHIPS 803-1419481 - Life Preserver, Vest, Inherently Buoyant: Pattern for Collar Section and Miscellaneous Details.
NAVSHIPS 803-1419482 - Life Preserver, Vest, Inherently Buoyant: Patterns for Inserts.

(Application for copies should be addressed to: Commander, Portsmouth Naval Shipyard, Code 280.6, Portsmouth, NH 03801.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DoDISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM B 117 - Standard Practice for Operating Salt Spray (Fog) Apparatus. (DOD adopted)

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

2.4 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.2.

3.2 Material. The materials shall be as specified herein and in accordance with the applicable drawings.

3.2.1 Drill. Drill shall conform to type I, class 3 in accordance with CCC-C-426 (see 4.5.2). The color shall conform to gray (color number 36076) or sea rescue orange (color number 12246) in accordance with FED-STD-595, as specified (see 6.2).

3.2.2 Inherently buoyant foam. The inherently buoyant foam shall conform to MIL-P-15280, form "S" sheet.

3.2.3 Webbing. Webbing shall be 1-inch wide nylon webbing conforming to MIL-W-17337, class 1. The webbing shall be equal in color to the cotton drill.

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3.2.4 Tape. Tapes shall conform to type I, class 3, in accordance with A-A-52071 except that the weight and breaking strength of tie loops, collar tie tapes, jacket tie tapes, and tunnel tapes shall be as specified in 3.2.4.2 and 3.2.4.3. Tapes shall be of the lengths indicated on the applicable drawings. Tapes shall be equal in color to the cotton drill.

3.2.4.1 Binding tape. The binding tape shall be made from 3/4-inch tape having a breaking strength of not less than 65 pounds.

3.2.4.2 Tie loops and collar tie tapes. Tie loops and collar tie tapes shall be made from 3/4-inch tape weighing not less than 0.18 ounce per linear yard and having a breaking strength of not less than 120 pounds.

3.2.4.3 Jacket tie tapes and tunnel tapes. Jacket tie tapes and tunnel tapes shall be made from 1-1/8 inch cotton tape weighing not less than 0.27 ounce per linear yard and having a breaking strength of not less than 180 pounds.

3.2.5 Retroreflective tape. Tape shall be a stitch-on type retroreflective tape conforming to 46 CFR 164.018, type I. Unless otherwise specified (see 6.2), the tape shall only be used on SEA Rescue Orange Preservers (color number 12246 in accordance with FED-STD-595).

3.2.6 Metal hardware. Strap tips, D-rings, adjustable tension slides, and snaphooks shall be in accordance with the applicable drawings. The metal hardware shall be made from brass or corrosion resistant steel and shall have a dull black finish. The metal hardware shall pass the salt spray test specified in 4.5.4.

3.2.7 Slide fasteners. Slide fasteners shall be nylon or polyester on nylon or polyester tape, and conform to type I, style 2, size LM, in accordance with V-F-106, except that the tape width shall be 9/16 inch. The length of slide fasteners shall be 10-1/4 inches. Fasteners shall pass the salt spray test specified in V-F-106. The color of the tape shall match the color of the cotton drill specified (see 3.2.1). Metal components of the slide fastener shall be of the composition specified in V-F-106.

3.2.8 Thread. Thread used in stitching of the preserver body shall conform to V-T-285, type II, class 1, size "E" polyester thread. Thread color and color- fastness shall be equal to that of the cotton drill. Thread used in stitching the nylon webbing to the preserver body and for webbing attachments shall conform to V-T-295, type II, size "E", and shall be colored to match the webbing.

3.2.9 Adhesive. Adhesive used for bonding the buoyant foam pads shall conform to MIL-A-24179, type I (see 3.3.4).

3.3 Design. The life preserver shall consist of an outside and inside jacket section, collar (except on type II), body strap, leg straps, and buoyant foam pads.

3.3.1 Outside and inside jacket sections. The outside and inside jacket sections shall be made from cotton drill (see 3.2.1).

3.3.2 Collar. The collar for the type I preserver shall be made from cotton drill (see 3.2.1).

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3.3.3 Body strap and leg straps. Body strap and leg straps shall be made of 1-inch nylon webbing (see 3.2.3) in accordance with the applicable drawings in section 2. The body strap assembly shall have a breaking strength of not less than 360 pounds when tested as specified in 4.5.5.

3.3.4 Inherently buoyant foam pads. There shall be four front pads (one upper and one lower pad on each side of the front) and one back pad. Type I, III life preservers shall also have one collar pad. The pads for the front and back may be made from a maximum of three plies of foam permanently bonded together equaling 2-1/2 inches in thickness.

3.3.4.1 Distribution of buoyant medium. The buoyant foam pads shall be cut in accordance with Drawing 803-1419482.

3.3.4.2 Buoyancy. The pads shall have the buoyancies specified in table I when tested as specified in 4.5.6.

TABLE I. Buoyancy.

Pad	Buoyancy (pounds)
Back pad	7.0 ± 0.5
Upper front pad	4.0 ± 0.5
Lower front pad	7.5 ± 0.5

3.3.5 Inspection label. A label conforming to DDD-L-20, type VI, class 1, 2-1/2 by 3-1/2 inches, shall be sewed on the bottom of the jacket on the inside of the lining. The printing on the tag shall be in waterproof black ink and shall be in accordance with Drawing 803-1419479.

3.4 Stitches, seams, and stitching. Stitches shall be type 301 in accor- dance with FED-STD-751 and shall be 8 to 10 stitches per inch. Seam types and stitching types shall conform to the applicable drawings of FED-STD-751, unless otherwise specified (see 6.2). Reflective tape stitching shall be around the outer perimeter of the tape, 3/16 ± 1/16 inch in from edge of tape.

3.5 Construction. The manufacture and assembly of type I and type II life preservers shall be in accordance with Drawings 803-1419479, 803-1419480, 803-1419481, and 803-1419482, as applicable. Actual construction shall be in accordance with the operations shown in table II, for type I preservers and table III for type II preservers. The contractor is not required to follow the exact sequence of operations, provided that the finished items are identical to those produced by following the sequence as listed in tables II and III.

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TABLE II. Construction of type I preserver.

No.	Operation	Stch type	Seam stch type	Stch inch
	NOTE: Order of sequences may not necessarily be in order listed.			
1	Cut parts I/A/W applicable drawings: A. Inside jacket section. B. Outside jacket section. C. Collar section. D. Slider protection patch. E. Unicellular foam as required.			
2	Cut 2 pieces webbing 3/4 inch wide X 2 inches long (turn 1/4 inch X 1/4 inch and hem one end) Weave D-ring into assembly and stitch to outer jacket on both sides I/A/W premarked locations.	301	EFb	8/10
3	Cut 2 pieces webbing 1-1/8 inches wide X 12 inches long and stitch 1/8 inch from edge and back lock stitch to outer jacket fronts on both sides.	301	Efb	8/10
4	Cut 2 pieces tie tapes 3/4 inch wide X 12 inches long attach to collar I/A/W designated marks on both ends of collar.	301	EFb	8/10
5	Cut 2-inch width reflective tapes: A. 4 inches long - 4 each. B. 12 inches long - 2 each. C. 3 inches long - 1 each. D. 7-1/2 inches long - 2 each.			
6	Stitch reflective tape to front and back of outside jacket sections I/A/W premarked locations from drawings - 7 places 3/16 inch on all four sides.	301	SSa	8/10
7	Stitch reflective tapes to front of collar on outside section I/A/W premarked locations from drawing - 2 places 3/16 inch on all four sides.	301	SSa	8/10

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TABLE II. Construction of type I preserver - Continued.

No.	Operation	Stch type	Seam stch type	Stch inch
8	Cut webbing 1 inch wide X 4-1/4 inches long for light attachment loop, fold in half, stitch ends at fold, and box-x to left side of outer section of front jacket.	301	SSat	Box-X
9	Stitch label to premarked location on outside of the inside jacket section 1/8 inch on all four sides	301	SSa	8/10
10	Bind slits (2 each) of outside jacket section with 3/4 inch wide tape.	301	SSa	8/10
11	Join armholes in and out jacket sections:			
	A. With both inside and outside sections (inner surfaces together) stitch 1/2 inch from edge around perimeter of both armholes.	301	SSa	8/10
	B. Open flat 1/2 inch seams and stitch 1 inch wide tape using double needle 1/8 inch from edges.	301	Ssad	8/10
12	Join shoulder seams:			
	A. Inner surfaces together join both shoulder seams 1/2 inch from edge.	301	SSa	8/10
	B. Stitch front starting approx-imately 3/4 inch from shoulder seam.			
13	Closing front of vest (both sides). From shoulder seam starting 3/4 inch down stitch 1/2 inch seam, taking in D-ring and tie tape assemblies at premarked notches.	301	Lsb	8/10

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TABLE II. Construction of type I preserver - Continued.

No.	Operation	Stch type	Seam stch type	Stch inch
14	Installing left and right side slide fasteners:			
	A. Place slide fasteners (open position) from bottom. Turn under fabric 1/2 inch and stitch 1/4 inch from folded edge (right side).			
	B. Repeat on opposite side (left).	301	LSb	8/10
	C. S/N reinforcement stitches to top and bottom both slide fasteners.	301		8/10
15	Slide protective patches:			
	A. Protective patch cut from base fabric with top edge on selvage.			
	B. Fold under remaining three sides and top stitch at bottom of slide I/A/W premarked locations.	301	LSd	8/10
16	Assemble the collar section:			
	A. Both collar pieces (right sides to right side) sew 1/2 inch from edge at neck opening of collar to premarked locations, leaving slide fastener opening, back tacking 1/2 inch starting and stopping, skip over slide fastener opening and complete other side in same manner.	301	SSa	8/10
	B. Turn collar and top stitch 1/8 inch over tie tapes on both sides.	301	SSa	8/10
	C. Set slide fastener in collar opening place protective patch over slide fastener end and stitch 1/8 inch from edge.	301	Ssa	8/10
	NOTE: Slide fastener when closed is to lay with slide fastener tongue to right of collar.			

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TABLE II. Construction of type I preserver - Continued.

No.	Operation	Stch type	Seam stch type	Stch inch
17	Set collar in neck opening (slide fastener tongue closed will finish on right side):			
	A. Place collar between front and back sections of vest, stitching through all four pieces, approximately 4-1/2 inches of front to shoulder section. Continue stitching with two ply for length of front and back of collar of 4 pieces or plies of fabric approximately 4-1/2 inches of neck section.	301	SSa	8/10
	B. Single needle stitch on collar I/A/W premarked locations forming a retaining pocket when foam is inserted.	301	SSa	8/10
	C. Bartack 1/2 inch both edges of above stitch patterns.	301	Bartack	8/10
18	Top stitch rest of fronts 1/8 inch from edges.	301	SSc	8/10
19	Slot for large D-ring:			
	A. Cut slit I/A/W drawing and location to inside and outside jacket sections.			8/10
	B. Fold four edges 3/8 inch inward to form a 3/4 inch X 3-3/4 inches rectangular opening through both inner and outer jacket.			8/10
	C. S/N four sides (4 plies) 1/16 inch from edges.	301	SSc	14/20
	D. S/N angle tack stitches in all four corners for reinforcement.	301	SSa	6/8
20	S/N stitch across back above D-ring slot I/A/W premarked locations, stitching inner and outer jacket together.	301	SSa	8/10

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TABLE II. Construction of type I preserver - Continued.

No.	Operation	Stch type	Seam stch type	Stch inch
21	S/N stitch starting under armhole along premarked locations curving toward front, following marks at approximately 3 inches from bottom across left front, the back and front along curved marking, ending under eight armhole barlock stitch both ends.	301	SSa	8/10
22	Tunnel tapes (2 each):			
	A. Cut tunnel tapes, webbing 1-1/8 inches wide X 66 inches long - 2 each.			
	B. Stitch end of one webbing to jacket at edge of tunnel opening with a diamond shape box-X 1 inch X 1 inch to inner side of outer jacket.	301	Box stitch -X	8/10
	C. Repeat above with 2nd webbing Box-X, 1 inch X 1 inch on opposite end of tunnel to be formed.	301	Box stitch -X	8/10
23	Close tunnel:			
	A. Close tunnel at bottom of entire jacket using 3/4 inch wide binding tape with both inner and outer jacket sections together, S/N 1/8 inch from edges of webbing.	301	BSa	8/10
	B. Fold ends of binding tape and insert in opening at both ends of vest and bartack.	301	Bartack	
24	Bartack upper section of shoulders near slide fastener on back section (2 places).		Bartack	

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TABLE II. Construction of type I preserver - Continued.

No.	Operation	Stch type	Seam stch type	Stch inch
25	Chest strap (1 each): A. Cut webbing 1 inch X 63 inches long. B. Turn under 2 inches forming loop for D-ring and Box-X 3/4 inch X 1-1/4 inches on left end of webbing. C. On opposite end (right side) weave webbing with 1 inch slide and snaphook turn or fold end to form loop 2-1/2 inches and boxstitch-X, 3/4 inch X 3/4 inch.	301 301	Box-X Box-X	8/10 8/10
26	Front leg straps (2 each): A. Cut webbing 1 inch X 7 inches (2 each). B. Fold under one end of webbing 7/8 inch then form a loop with 2 each D-rings inserted inside allowing both ends of webbing to meet. C. Stitch both front leg straps to premarked areas with box-X stitch 3/4 inch X 1-1/2 inches.	301	Box stitch -X	8/10
27	Back leg straps (2 each): A. Cut webbing 1 inch X 48 inches (2 each). B. Press ball-tip to each 48 inches webbing. C. Fold opposite end of webbing 7/8 inch on itself and stitch to premarked locations with boxstitch-X, 3/4 inch X 1-1/2 inches.	301	Box stitch -X	8/10
28	Visual inspections: A. Thread breaks, skips, loose ends. B. General appearance.			

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TABLE III. Construction of type II preserver.

No.	Operation	Stch type	Seam Stch type	Stch Inch
	NOTE: Order of sequences may not necessarily be in order listed.			
1	Cut fabric parts I/A/W applicable drawings: A. Inside jacket section. B. Outside jacket section. C. Chafing patch. D. Slider protection patch as required.			
2	Cut 2-inch width reflective tape: A. 4 inches long - 2 each. B. 12 inches long - 2 each. C. 3 inches long - 1 each. D. 7-1/2 inches long - 2 each.			
3	Stitch reflective tape to front and back of outside jacket section I/A/W premarked locations from drawings, seven places, 3/16 inch on all four sides.	301	SSa	8/10
4	Bind slits of outside jacket section with 3/4 inch width tape.	301	BSa	
5	Stitch label to premarked location on outside of the inside jacket section 1/8 inch on all four sides.	301	SSa	8/10
6	Join armholes in and out jacket sections: A. With both inside and outside sections (inner surfaces together) stitch 1/2 inch from edge around perimeter of both armholes. B. Open flat 1/2 inch seams and stitch 1 inch width tape using double needle 1/8 inch from edges.	301 301	SSa SSad	8/10 8/10
7	Join shoulder seams: A. Inner surfaces together join both shoulder seams 1/2 inch from edge.	301	SSa	8/10

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TABLE III. Construction of type II preserver - Continued.

No.	Operation	Stch type	Seam stch type	Stch inch
	B. Stitch front starting 3/4 inch from shoulder seam.	301	SSa	8/10
8	Cut 2 pieces webbing 1-1/8 inches wide X 12 inches long: A. Turn one end doubled 1/4 inch and stitch 1/8 inch from edge and back lock stitch.	301	EFb	8/10
9	Closing front of vest, both sides: A. From shoulder seam starting 3/4 inch down, stitch 1/2 inch seam to premarked notch.	301	SSa	8/10
	B. Insert 1-1/8 inches X 12 inches jacket tie tape, continue to stitch to 2nd notch (backtack locking stitch), leave opening for installing slide fastener.	301	SSj	8/10
10	Installing left and right side slide fasteners: A. Place slide fastener (open position) from bottom. Turn under fabric 1/2 inch and stitch 1/4 inch from folded edge (right side).	301	LSb	8/10
	B. Repeat on opposite (left side).	301	LSb	8/10
	C. S/N reinforcing stitches to top and bottom of both slide fasteners.	301		8/10
11	Slider protective patch: A. Protective patch cut from base fabric with top edge on selvage.			
	B. Fold under remaining three sides 1/4 inch and topstitch at bottom of slide fastener I/A/W premarked locations.	301	LSd	8/10
12	Installing neck slide fastener: A. Stitch fastener to neck area with opening to close when slide fastener is in closed position on right side.	301	Lsb	8/10

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TABLE III. Construction of type II preserver - Continued.

No.	Operation	Stch type	Seam stch type	Stch inch
	Fold fabric 1/2 inch and S/N 1/4 inch from edge, add stitch to slide fastener half at edge to form an easy to work with operation.			
	B. Cut chafing patch (3-1/2 X 12 inches) 1 each turn under ends 3/8 inch (DO NOT STITCH).	301	SSc	8/10
13	Top stitch rest of fronts 1/8 inch from edges.	301	SSc	8/10
14	Slot for large D-ring:			
	A. Cut slit I/A/W drawing and location to inside and outside jacket sections.			
	B. Fold four edges 3/8 inch inward to form a 3/4 inch X 3-3/4 inch rectangular opening through both inner/outer jacket.			
	C. S/N four sides (4 plies) 1/16 inch from edges.	301	SSc	8/10
	D. S/N angle stitches in all four corners for reinforcement.	301	SSa	8/10
15	S/N stitch across back above D-ring slot I/A/W premarked locations, stitching inner and outer jacket together.	301	SSa	8/10
16	S/N stitch starting under armhole along premarked locations curving toward front, following marks at approximately 3 inches from bottom, across left front, the back and right front along curved marking, ending under right armhole backlock stitch both ends.	301	SSa	8/10
17	Tunnel tapes (2 each):			
	A. Cut tunnel tapes, webbing 1-1/8 inches width X 66 inches long - 2 each.			

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TABLE III. Construction of type II preserver - Continued.

No.	Operation	Stch type	Seam stch type	Stch inch
	B. Stitch end of one webbing to jacket at edge of tunnel opening with a diamond shape box-x, 1 inch X 1 inch to inner side of outer jacket.	301	Box-stitch X	8/10
	C. Repeat above with 2nd webbing box-x, 1 inch X 1 inch on opposite end of tunnel to be formed.	301	Box-stitch X	8/10
18	Close tunnel:			
	A. Close tunnel at bottom of entire jacket using 3/4 inch width binding tape with both inner and outer jacket sections together, S/N 1/8 inch from edges of webbing.	301	BSa	8/10
	B. Fold ends of binding tape and insert in opening at both ends of vest and bartack.		Bartack	
19	Bartack upper section of shoulders near slide fastener on back section (2 places).		Bartack	
20	Chest strap (1 each):			
	A. Cut webbing 1 inch X 63 inches long.			
	B. Turn under 2 inches forming loop for D-ring and box-x 3/4 inch X 1-1/4 inches on left end of webbing.	301	Box-X	
	C. On opposite end (right side) weave webbing with 1 inch slide and snaphook, turn or fold end to form loop 2-1/2 inches and boxstitch X, 3/4 inch X 3/4 inch.	301	Box-X	8/10
21	Front leg straps (2 each):			
	A. Cut webbing 1 inch X 7 inches (2 each).			
	B. Fold under one end of webbing 7/8 inch then form a loop with 2 each D-rings inserted inside allowing both ends webbing to meet.			

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TABLE III. Construction of type II preserver - Continued.

No.	Operation	Stch type	Seam stch type	Stch inch
	C. Stitch-both front leg straps to premarked areas with box-x stitch 3/4 inch X 1-1/2 inches.	301	Box-stitch X	8/10
22	Back leg straps (2 each): A. Cut webbing 1 inch X 48 inches (2 each). B. Press ball-tip to each 48-inch webbing. C. Fold opposite end of webbing 7/8 inch on itself and stitch to premarked locations with boxstitch X, 3/4 inch X 1-1/2 inch.	301	Box-stitch X	8/10
23	Visual inspections: A. Thread breaks, skips, loose ends. B. General appearance.			

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.2).
(b) Conformance inspection (see 4.3).

4.2 First article inspection. First article inspection shall be performed on the submitted samples as specified in 3.1 when specified in 6.2. First article inspection shall consist of the examination and tests specified in table IV.

4.2.1 Samples for first article inspection. Before starting regular production, the contractor shall submit the following samples for first article inspection:

(a) Two complete life preservers.
(b) Two linear yards (full width) of each type fabric used.
(c) Four yards of each type of tape and webbing used.
(d) Two complete sets of buoyant pads.
(e) Two complete sets of all hardware used.
(f) One cone of each type of thread used.
(g) Six slide fasteners.
(h) Two complete body strap assemblies.

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4.2.2 First article sample identification. First article samples shall be forwarded to the Commander, Naval Surface Warfare Center, Carderock Division, Annapolis Detachment, 3A Leggett Circle, Attn: Code 643, Annapolis, MD 21402-5067. The samples shall be plainly identified by securely attached durable tags marked with the following information:

(a) "Samples for first article tests."
(b) Name of product, together with complete identifying data, such as manufacturer's designation number.
(c) Name of manufacturer.
(d) Contract or purchase order number.

4.3 Conformance inspection. Conformance inspection shall be conducted as specified in table IV.

TABLE IV. First article and conformance inspection.

Test	Require-ment	Test	First article	Conformance
Group A:				
Material	3.2	4.4	X	X
Design	3.3	4.4	X	X
Stitching	3.5	4.4	X	X
Construction	3.4	4.4	X	X
Group B:				
Drill	3.2.1	4.5.2	X	
Webbing	3.2.3	4.5.3	X	
Binding tape	3.2.4.1	4.5.3	X	
Tie loops and collar tie tapes	3.2.4.2	4.5.3	X	
Tie tapes and tunnel tapes	3.2.4.3	4.5.3	X	
Salt spray	3.2.6	4.5.4	X	
Breaking strength of body strap assembly	3.3.3	4.5.5	X	X
Buoyancy	3.3.4.2	4.5.6	X	X

4.3.1 Inspection lot. For the purpose of conformance inspection and test sampling, a lot is defined as all the completed life preservers of the same type, with body strap assemblies, produced in one facility, using the same production processes and materials, and being offered for delivery at one time. A lot shall consist of not greater than 3200 life preservers.

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4.3.2 Sampling for conformance group A inspection. As a minimum, the contractor shall randomly select a sample quantity from each lot of completed life preservers as specified in table V and inspect them as specified in table IV for the defects listed in table VII (see 6.4).

TABLE V. Lot sampling for group A inspection.

Lot size	Critical	Major	Minor
2 to 8	All	All	3
9 to 25	All	13	3
26 to 50	All	13	5
51 to 90	All	13	6
91 to 150	All	13	7
151 to 280	All	20	10
281 to 500	All	29	11
501 to 1200	All	34	15
1201 to 3200	All	42	18

4.3.3 Sampling for conformance group B tests. As a minimum, the contractor shall randomly select a sample quantity of completed life preservers as specified in table VI and subject them to the tests as specified in table IV (see 6.4).

TABLE VI. Sampling for group B tests.

Lot size	Body strap assembly	Buoyancy
2 to 15	3	2
16 to 25	3	3
26 to 50	5	5
51 to 90	6	5
91 to 150	7	6
151 to 280	10	7
281 to 500	11	9
501 to 1200	15	11
1201 to 3200	18	13

4.4 Visual and dimensional examination. Sample life preservers, complete with body strap assemblies, selected as specified in 4.3.2 shall be thoroughly examined to determine conformance with the requirements of this specification. Examination shall be conducted as specified in table IV for conformance to the criteria of table VII.

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TABLE VII. Classification of defects.

Categories	Defects
Critical: 1	Components damaged, incorrect, missing, misfitting, or faulty, impairing the function of the preserver as life saving equipment.
Major: 101 102 103 104 105	Positioning of body strap assembly and leg straps is incorrect. Stitching of cotton drill, webbing, and tapes, nonconforming; not correct type stitch. Slider protection patch, chafing patch, located as specified. Dimensions not as specified on applicable drawings. Bond failure of buoyant foam pads, if applicable.
Minor: 201 202 203	Completed preserver not clean; evidence of dirt or oil spots. Improper tensioning in stitching in parts other than the body strap assembly or leg straps. Incorrect or illegible marking or both of preserver.

4.5 Tests.

4.5.1 General. All tests not herein specified shall be in accordance with the methods specified in FED-STD-191.

4.5.2 Drill. The drill shall be tested for conformance to CCC-C-426 using the following tests in accordance with FED-STD-191 to determine compliance with 3.2.1.

Test	Method
Weight	5041
Yarns/inch	5050
Breaking strength	5100

4.5.3 Webbing and tapes. All webbing and tapes shall be subjected to the following tests in accordance with FED-STD-191 to determine compliance with 3.2.3 and 3.2.4.

Test	Method
Weight	5041
Breaking strength	4108

4.5.4 Salt spray. A 96-hour salt spray shall be conducted on all metal hardware and slide fasteners in accordance with ASTM B 117. There shall be no evidence of corrosion which will adversely affect the operation of the equipment.

4.5.5 Breaking strength of body strap assembly. The body strap assemblies shall be tested for conformance to 3.3.3, using one of the following methods:

4.5.5.1 Testing machine method. A section of the strap assembly with the D-ring engaged in the snap hook and including the sliding buckle shall be taken for the test. The specimen shall be of sufficient length so that the distance between the jaws of the testing machine shall be approximately 10 inches after the specimen has been clamped for testing. The rate of separation of the jaws of the testing machine shall be approximately 12 inches per minute. The strap assembly shall have a breaking strength of not less than 360 pounds.

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4.5.5.2 Alternate dead weight method. A portion of the front body strap assembly, with the D-ring engaged in the snap hook, and including the sliding buckle, shall be taken for the test. One end of this assembly shall be fastened to a weight of 360 pounds and the other end shall be attached to a suitable lifting device having a maximum rate of lift not greater than 12 inches per minute. After carefully taking up the slack, the weight shall be lifted free of its point of rest and allowed to swing clear for not less than 1 minute. The strap assembly shall not break during this test. Tests conducted by this method shall be made at prevailing atmospheric conditions.

4.5.6 Buoyancy. The completed front and back pads shall be individually tested to show compliance with 3.3.4.2. In this test, the pad shall be placed in a cage of adequate size and of sufficient weight to submerge the pad. This assembly shall be suspended from a direct reading scale, in fresh water, at room temperature, with the top of the cage 2 inches beneath the surface of the water. The scale reading shall be taken as soon as practicable and the buoyancy calculated as the apparent weight of the cage and pad subtracted from the previously determined underwater weight of the assembly without the pad.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department or Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. Life preservers made in accordance with this specifi-cation are intended for use on surface ships.

6.2 Acquisition requirements. Acquisition documents must specify the following:

(a) Title, number, and date of this specification.
(b) Type required (see 1.2).
(c) Issue of DoDISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.2.1 and 2.3).
(d) When first article is required (see 3.1 and 4.2).
(e) Color of preserver required (see 3.2.1).
(f) When retroreflective tape is used other than as specified (see 3.2.5).
(g) When seam and stitching types are other than as specified (see 3.4).
(h) Packaging requirements (see 5.1).

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6.3 First article. When first article inspection is required, the contracting officer should provide specific guidance to offerors whether the item(s) should be a preproduction sample, a first article sample, a first production item, a sample selected from the first lot production items, a standard production item from the contractor's current inventory (see 3.1), and the number of items to be tested as specified in 4.3. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.4 Lot acceptance and rejection criteria. If one or more defects are found in any sample, the entire lot represented by the sample should be rejected. If a lot is rejected, the contractor has the option of screening 100 percent of the lot for the defective characteristic(s) or providing a new lot which is to be inspected in accordance with the sampling plan contained herein. The contractor should maintain for a period of three years after contract completion all records of inspections, tests, and any resulting rejections.

6.5 Subject term (key word) listing.

Collar
Foam pads
Outer jacket
Slide fastener

6.6 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodians:
Army - ME
Navy - SH
Air Force - 99

Preparing activity:
Navy - SH
(Project 4220-0392)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.

2. The submitter of this form must complete blocks 4, 5, 6, and 7.

3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	**1. DOCUMENT NUMBER** MIL-DTL-18045G	**2. DOCUMENT DATE (YYMMDD)** 970701

3. DOCUMENT TITLE LIFE PRESERVERS, VEST, INHERENTLY BUOYANT

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME *(Last, First, Middle Initial)*	b. ORGANIZATION	
c. ADDRESS *(Include Zip Code)*	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON *(if applicable)*	**7. DATE SUBMITTED** (YYMMDD)

8. PREPARING ACTIVITY

a. NAME A. THOMPSON	b. TELEPHONE *Include Area Code)* (1) Commercial (703) 602-0841 (2) AUTOVON 332-0841
c. ADDRESS *(Include Zip Code)* COMMANDER, SEA 03R42, NAVAL SEA SYSTEMS COMMAND, 2531 JEFFERSON DAVIS HIGHWAY ARLINGTON, VA 22242-5160	**IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:** DEFENSE QUALITY AND STANDARDIZATION OFFICE 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22401-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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